Linghao Kong

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Education

- **Massachusetts Institute of Technology** (MIT) — Cambridge, MA
  - PhD candidate in Electrical Engineering and Computer Science (GPA: 4.91/5.00) — *Sep 2022 – Present*
    - Advised by Professor Nir N. Shavit
  - SM in Electrical Engineering and Computer Science (GPA: 4.91/5.00) — *Sep 2022 – May 2024*
    - Advised by Professor Nir N. Shavit
    - Thesis: "Sparse Expansion and Neuronal Disentanglement"
- **Columbia University in the City of New York** (CU) — New York, NY
  - BA in Computer Science and in Neuroscience and Behavior (GPA: 3.97/4.00) — *Sep 2018 – May 2022*
    - Honors: *magna cum laude*, Dean's List for all semesters

Research Experience

- **PhD Candidate**, *Shavit Lab*, MIT — *Sep 2022 – Present*
  - Initiated and led research on feature-aware sparsity in LLMs, discovering and characterizing Wasserstein neurons: entangled neurons that map similar inputs to dissimilar outputs and are disproportionately fragile under pruning.
  - First demonstrated that negative pre-activations in smooth-activation LLMs contain structured syntactic computation, establishing a double dissociation between grammatical and non-grammatical capabilities.
  - Designed Wisp, Wisp+, and Whisper, a family of first- and second-order LLM pruning methods inspired by how Wasserstein neurons differentiate similar inputs, achieving state-of-the-art one-shot pruning performance.
  - Developed a broader feature-aware sparsity framework showing that sparse performance depends on preserving feature-space computation and reducing superposition-driven interference, not merely preserving large weights.
- **Research Intern**, *Machine Learning Research Team*, Red Hat — *May 2025 – Aug 2025*
  - Formulated an interpretable latency model for speculative decoding in LLM serving, extending speedup analysis beyond isolated settings and informing more effective draft model and draft length choices.
- **Research Intern**, *Machine Learning Research Team*, Neural Magic (acquired by Red Hat) — *Jun 2024 – Aug 2024*
  - Spearheaded FP8 LLM quantization initiative to reduce inference cost: primary contributor to top-8 trending, most extensive FP8 model collection on Hugging Face with 2M+ downloads, featured by NVIDIA and MarkTechPost.
- **Research Assistant**, *Sims Lab*, CU — *Jan 2019 – Aug 2022*
  - Used scGen and single-cell perturbation analysis to model cancer cell fate responses; contributed experimental and computational work that challenged a prevailing model of active translation.
- **Research Assistant**, *Laboratory for Fluorescence Dynamics*, UC Irvine — *Aug 2015 – Jul 2018*
  - Conducted multi-year cancer cell metabolism research, culminating in an independent project using fluorescence imaging and automated ImageJ analysis; modeled tumor spheroid growth in COSMOS summer research program.

Publications (* denotes equal contribution, † denotes co-correspondence)

- **Kong, L.***, Subramanian, I.*, Shavit, Y., Adler, M., Alistarh, D., & Shavit, N. N. (2026). **Expand neurons, not parameters**. *The 43rd International Conference on Machine Learning* (**ICML 2026**). https://arxiv.org/abs/2510.04500
- **Kong, L.**, Ning, A., Adler, M., & Shavit, N. N. (2026). **Negative pre-activations differentiate syntax**. *The 14th International Conference on Learning Representations* (**ICLR 2026**). https://openreview.net/forum?id=RzcCrU0tXP
- **Kong, L.**, Flynn, M., Peng, M., Shavit, N. N., Kurtz, M., & Marques, A. N. (2026). **An interpretable latency model for speculative decoding in LLM serving**. arXiv preprint. https://arxiv.org/abs/2605.15051
- **Kong, L.**, Subramanian, I., Adler, M., Alistarh, D., Gutfreund, D., & Shavit, N. N. (2026). **The sparsity whisperer**. Under review.
- **Kong, L.**, Adler, M., & Shavit, N. N. (2026). **The feature-space alignment hypothesis for neural network sparsity**. Under review.
- Tumma, N.*, **Kong, L.***†, Sawmya, S., Wang, T. T., & Shavit, N. N.† (2025). **A connectomics-driven analysis reveals novel characterization of border regions in mouse visual cortex**. *Neural Networks*, 190, 107688 (**Neural Netw**). https://doi.org/10.1016/j.neunet.2025.107688
- Sawmya, S.*, **Kong, L.***, Markov, I., Alistarh, D., & Shavit, N. N. (2025). **Wasserstein distances, neuronal entanglement, and sparsity**. *The 13th International Conference on Learning Representations* (**ICLR 2025, Spotlight Presentation**). https://openreview.net/pdf?id=cnKhHxN3xj
- Hobson, B. D., **Kong, L.**, Angelo, M. F., Lieberman, O. J., Mosharov, E. V., Herzog, E., Sulzer, D., & Sims, P. A. (2022). **Subcellular and regional localization of mRNA translation in midbrain dopamine neurons**. *Cell Reports*, 38(2) (**Cell Rep**). https://doi.org/10.1016/j.celrep.2021.110208

- Hobson, B. D., **Kong, L.**, Hartwick, E. W., Gonzalez, R. L., Jr., & Sims, P. A. (2020). **Elongation inhibitors do not prevent the release of puromycylated nascent polypeptide chains from ribosomes**. *eLife* 9, e60048 (**eLife**). https://doi.org/10.7554/eLife.60048
- **Kong, L.***, Murata, M. M.*, & Digman, M. A. (2018). **Absence of REV3L promotes p53-regulated cancer cell metabolism in cisplatin-treated lung carcinoma cells**. *Biochemical and Biophysical Research Communications*, *496*(1), 199-204 (**BBRC**). https://doi.org/10.1016/j.bbrc.2018.01.026

Conferences and Workshops (* denotes equal contribution)

- Liu, G., Shavit, N. N., & **Kong, L.** (2026, July). **Characterizing plastic regions in neural networks** [Poster presentation]. *The Workshop on Continual Adaptation at Scale: Towards Sustainable AI at the 43rd International Conference on Machine Learning* (**ICML 2026 Workshop CATS**), Seoul, South Korea.
- **Kong, L.**, Adler, M., & Shavit, N. N. (2026, April). **The feature-space alignment hypothesis for neural network sparsity** [Poster presentation]. *The 2nd Workshop on Scientific Methods for Understanding Deep Learning at the 14th International Conference on Learning Representations* (**ICLR 2026 Workshop Sci4DL**), Rio de Janeiro, RJ, Brazil.
- **Kong, L.***, Ning, A., & Shavit, N. N. (2025, July). **Input differentiation via negative computation** [Poster presentation]. *The 3rd Workshop on High-dimensional Learning Dynamics at the 42nd International Conference on Machine Learning* (**ICML 2025 Workshop HiLD**), Vancouver, BC, Canada.
- **Kong, L.***, Durresi, H., Mi, L., & Shavit, N. N. (2025, March). **Presynaptic input synchrony at scale** [Poster presentation]. *Computational and Systems Neuroscience* (**COSYNE 2025**), Montreal, QC, Canada.
- Sawmya, S.*, **Kong, L.***, Markov, I., Alistarh, D., & Shavit, N. N. (2024, August). **Neuronal disentanglement and Sparse Expansion** [Poster presentation]. *New England Mechanistic Interpretability Workshop Series* (**NEMI 2024**), Boston, MA, United States.
- Hobson, B. D., **Kong, L.**, Angelo, M. F., Lieberman, O. J., Mosharov, E. V., Herzog, E., Sulzer, D., & Sims, P. A. (2021, October). **Subcellular and regional localization of mRNA translation in midbrain dopamine neurons** [Poster presentation]. *2021 Columbia Undergraduate Research Symposium*, New York, NY, United States.
- Hobson, B. D., **Kong, L.**, Hartwick, E. W., Gonzalez, R. L., Jr., & Sims, P. A. (2020, October). **Elongation inhibitors do not prevent the release of puromycylated nascent polypeptide chains from ribosomes** [Poster presentation]. *2020 Columbia Undergraduate Research Symposium*, New York, NY, United States.
- **Kong, L.**, Hobson, B. D., & Sims, P. A. (2019, October). **Toward visualization of active translation in dopaminergic neurons** [Poster presentation]. *2019 Columbia Undergraduate Research Symposium*, New York, NY, United States.
- **Kong, L.***, Murata, M. M.*, & Digman, M. A. (2017, October). **Fighting the (chemotherapeutic) resistance: restoring p53 function and silencing REV3L suppresses the cancerous metabolic phenotype in cisplatin treated human non-small lung carcinoma cells** [Poster presentation]. *2nd World Congress on Cancer Research and Therapy* (**WCCRT 2017**), San Diego, CA, United States.

Honors and Achievements

- ICLR Official Travel Grant, MIT OGE Conference Grant for ICLR 2026 – Awardee *2026*
- IBM AI Research Grant – Primary author of funded proposal supporting my research *2025*
- Amazon AI PhD Fellowship – MIT nominee, results pending *2025*
- Apple Scholars in AI/ML PhD Fellowship – One of three MIT nominees *2025*
- Spotlight Presentation at ICLR – Top 5% of submissions *2025*
- Pillar VC Travel Grant for ICML 2025 – Awardee *2025*
- COSYNE New Attendee Travel Grant for COSYNE 2025 – Awardee *2025*
- Cerebras Research Fellowship – Awardee *2024*
- NSF Graduate Research Fellowship Program – Honorable Mention *2024*
- Columbia University I.I. Rabi Scholar – One of 17 students in class awarded yearly research funding *2018 – 2022*
- American Invitational Mathematics Examination (AIME) qualifier – Top 5% nationally *2015 – 2018*
- USA Biology Olympiad (USABO) semifinalist – Top 10% nationally *2015*

Teaching Experience

- **Teaching Fellow**, Department of Electrical Engineering and Computer Science, MIT *Sep 2025 – Dec 2025*
  - Developing problem sets and holding weekly office hours for the following courses:
  - 6.4610 Natural Language Processing *Fall 2025*
- **Research Mentor,** Department of Electrical Engineering and Computer Science, MIT *Dec 2022 – Present*
  - Mentoring undergraduate students to conduct research in mechanistic interpretability, model efficiency, and connectomics. Graduated mentees:
  - Heidi Durresi, now PhD student at MIT *Sep 2023 – Sep 2024*
  - Neehal Tumma, now PhD student at MIT *Dec 2022 – May 2024*
- **Course Assistant**, Computer Science Department, CU *Jan 2021 – May 2022*

- Guided students to implement the mathematical and theoretical principles taught in class in Python-based applications and problem sets through weekly office hours and biweekly lab sessions in the following courses:
- COMS W4701 Artificial Intelligence *Spring 2022*
- COMS W4733 Computational Aspects of Robotics *Fall 2021*
- COMS W4701 Artificial Intelligence *Summer 2021*
- COMS W3251 Computational Linear Algebra *Spring 2021*

❖ **Vice President**, *Orange County Math Circle* (OCMC), Orange County, CA *Nov 2013 – May 2018*
- Oversaw logistics of all other math clubs within OCMC; resolved club issues in weekly diagnostic meetings.
- Directed volunteers to serve 2800+ students yearly; trained others to better instruct students.

Invited Talks

❖ Talk on *Negative Pre-activations Differentiate Syntax*, MIT, Cambridge, MA, USA *Apr 2026*

❖ Talk on *Wasserstein Distances, Neuronal Entanglement, and Sparsity*, Red Hat, Boston, MA, USA *Mar 2025*

Extracurricular Activities

❖ **Editor in Chief**, *Columbia Science Review*, CU *Sep 2018 – May 2022*
- Supervised over 40 writers and editors to ensure the smooth operation of an online and a biannual publication.
- Coordinated between different teams, including illustrators and layout designers, to produce a cohesive product.

❖ **Vice President**, *Columbia Synapse*, CU *Sep 2019 – May 2021*
- Oversaw the organization of events to help unite the community in support of those with traumatic brain injuries, including research panels, socials, as well as of a large conference held during March 13th and 14th, 2021.

❖ **RASC-AL Mission Member**, *Columbia Space Initiative*, CU *Nov 2018 – May 2020*
- Semifinalist for NASA's RASC-AL competition to design a lunar lander, focused on thermal management.